

- AI-Based Student Performance Analytics System Scenario: Intelligent Education. A university wants to develop an AI-assisted Student Performance Analytics System to monitor student academic performance and identify students who may need additional academic support.
- Create a class `Student` with attributes such as Student ID, Name, Marks in different subjects, Average, and Performance Level.
 - Implement member functions to input student details, calculate average marks, determine performance level, and display the student report.
 - Use an array of objects to manage records of multiple students.
 - Include a static member to maintain the total number of students analyzed.
 - Apply proper encapsulation using private data members and public member functions.

Your Code

```
1 #include <iostream>
2 using namespace std;
3 class student {
4     int id,m[3];
5     float avg;
6     static int total;
7     public:
8     void input(){
9         cin>>id;
10        for(int i=0;i<3;i++)cin>>m[i];
11        total++;
12    }
13    void calculate(){
14        avg=(m[0]+m[1]+m[2])/3.0;
15    }
16    void display(){
17        cout<<"ID:"<<id<<"\naverage:"<<avg;
18        cout<<"\nPerformance:";
19        if(avg>=80)cout<<"excellent";
20        else if(avg>=60)cout<<"good";
21        else if (avg>=40)cout<<"average";
22        else cout<<"needs improvement";
23        cout<<"\n";
24    }
25    static void showtotal(){
26        cout<<"Total students:"<<total;
27    }
28 };
29 int student::total=0;
30 int main(){
31     int n;
32     cin>>n;
33     student s[50];
34     for(int i=0;i<n;i++){
35         s[i].input();
36         s[i].calculate();
37     }
38     for(int i=0;i<n;i++)
39         s[i].display();
40     student::showtotal();
41 }
```

Input

```
2
101
85 90 80
102
```

Output



Q2 Digital Healthcare Patient Monitoring System

10 marks

Digital Healthcare Patient Monitoring System Scenario: Smart Healthcare. A modern hospital wants to implement a Digital Patient Monitoring System that maintains patient information and automatically calculates basic health-risk indicators from recorded medical parameters.

- A. Design a class Patient with attributes such as Patient ID, Name, Age, Heart Rate, Blood Pressure, and Health Status.
- B. Implement member functions to input patient details, calculate/identify the health status based on predefined conditions, and display the patient report.
- C. Store multiple patient records using an array of objects.
- D. Use a static member to track the total number of patients monitored.
- E. Demonstrate data hiding, encapsulation, and structured object-oriented program design.

Your Code

```
1 #include <iostream>
2 using namespace std;
3 class patient {
4     int id,age,hr,bp;
5     string name,status;
6     static int count;
7 public:
8     void input(){
9         cin>>id>>name>>age>>hr>>bp;
10        count++;
11    }
12    void checkhealth(){
13        if(hr>=60&&hr<=100&&bp>=90&&bp<=120)
14            status="normal";
15        else
16            status="at risk";
17    }
18    void display(){
19        cout<<id<<" "<<name<<" "<<age<<" "<<hr<<" "<<bp<<" "<<status<<endl;
20    }
21    static int total(){
22        return count;
23    }
24 };
25 int patient::count=0;
26 int main(){
27     int n;
28     cin>>n;
29     patient p[10];
30     for(int i=0;i<n;i++){
31         p[i].input();
32         p[i].checkhealth();
33     }
34     for(int i=0;i<n;i++)
35         p[i].display();
36     cout<<"total patients:"<<endl;
37     patient::total();
38     return 0;
39 }
```

Input

```
2
101
ravi
25 80 110
```

Output



Q3 Digital Wallet and FinTech Transaction System

10 marks

Digital Wallet and FinTech Transaction System Scenario: Cashless Economy. A FinTech company wants to develop a Digital Wallet Management System to maintain customer wallet information and perform secure financial transactions.

- A. Create a class DigitalWallet with attributes such as Wallet ID, Customer Name, Mobile Number, and Balance.
- B. Implement member functions for wallet creation, money deposit, money withdrawal, balance enquiry, and transaction display.
- C. Use an array of objects to manage multiple customer wallets.
- D. Include a static member to maintain the total number of active wallets.
- E. Apply appropriate access specifiers and encapsulation to prevent direct modification of sensitive financial data.

Your Code

```
1 #include<iostream>
2 using namespace std;
3 class digitalwallet{
4     int walletid;
5     string customername,mobilenumber;
6     double balance;
7     static int activewallet;
8 public:
9     void createwallet(int id,string name,string mobile,double amount){
10         walletid=id;
11         customername=name;
12         mobilenumber=mobile;
13         balance=amount;
14         activewallet++;
15     }
16     void deposit(double amount){
17         balance+=amount;
18         cout<<"Money deposited successfully.\n";
19     }
20     void withdraw(double amount){
21         if(amount<=balance){
22             balance-=amount;
23             cout<<"money withdrawn sucessfully.\n";
24         }else{
25             cout<<"insufficient balance.\n";
```

Input

stdin input

Output

Q4 Smart Agriculture Crop Monitoring System

10 marks

Smart Agriculture Crop Monitoring System ♦ Scenario: IoT-Based Agriculture. An agricultural research centre is developing an IoT-based Smart Agriculture System to monitor crop fields using sensor data and identify whether crops require irrigation.

- A. Design a class CropField with attributes such as Field ID, Crop Name, Soil Moisture, Temperature, Humidity, and Irrigation Status.
- B. Implement member functions to input sensor readings, evaluate soil conditions, determine irrigation requirements, and display field information.
- C. Use an array of objects to maintain information about multiple crop fields.
- D. Include a static member to count the total number of monitored fields.
- E. Ensure proper encapsulation and modular object-oriented design suitable for an IoT-based agricultural application.

Your Code

```
1 #include<iostream>
2 using namespace std;
3 class cropfield{
4     private:
5         int fieldid;
6         string cropname;
7         float soilmoisture,temperature,humidity;
8         string irrigationstatus;
9         static int count;
10    public:
11        cropfield(){
12            count++;
13        }
14        void input(){
15            cin>>fieldid>>cropname>>soilmoisture>>temperature>>humidity;
16        }
17        void evaluate(){
18            if(soilmoisture<40)
19                irrigationstatus="required";
20            else
21                irrigationstatus="not required";
```

Input

```
2
101 Rice 35 30 70
102 Wheat 55 28 65
```

Output



Smart City IoT Energy Management System Scenario: Sustainable Smart City. A smart-city research project aims to develop an IoT-based Energy Management System for monitoring electricity consumption in different buildings and identifying buildings with high energy usage.

A. Create a class Building with attributes such as Building ID, Building Name, Building Type, Energy Consumption, and Energy Status.
B. Implement member functions to input building details, calculate energy status based on consumption, and display energy usage information.

C. Use an array of objects to manage multiple buildings in the smart-city system.

D. Include a static member to maintain the total number of buildings monitored.

E. Apply encapsulation, data hiding, and structured C++ object-oriented programming principles in the implementation.

Your Code

```
1 #include<iostream>
2 using namespace std;
3 class building{
4     private:
5         int id;
6         string name,type,status;
7         float consumption;
8         static int count;
9     public:
10    void input(){
11        cin>>id>>name>>type>>consumption;
12        count++;
13    }
14    void calculatestatus(){
15        if(consumption>500)
16            status="High";
17        else
18            status="Normal";
19    }
20    void display(){
21        cout<<"Building ID:"<<id<<endl;
22        cout<<"Building Name:"<<name<<endl;
23        cout<<"Building Type:"<<type<<endl;
24        cout<<"Energy Consumption:"<<consumption<<endl;
25        cout<<"Energy Status:"<<status<<endl;
26    }
27    static void showcount(){
28        cout<<"Total Buildings:"<<count<<endl;
29    }
30 };
31 int building::count=0;
32 int main(){
33     int n;
34     cin>>n;
35     building b[10];
36     for(int i=0;i<n;i++){
37         b[i].input();
38         b[i].calculatestatus();
39     }
40     for(int i=0;i<n;i++){
41         b[i].display();
42     }
43     building::showcount();
44     return 0;
```

Input

```
2
101 Hospital Public 650
102 School Educational 350
```

Output

